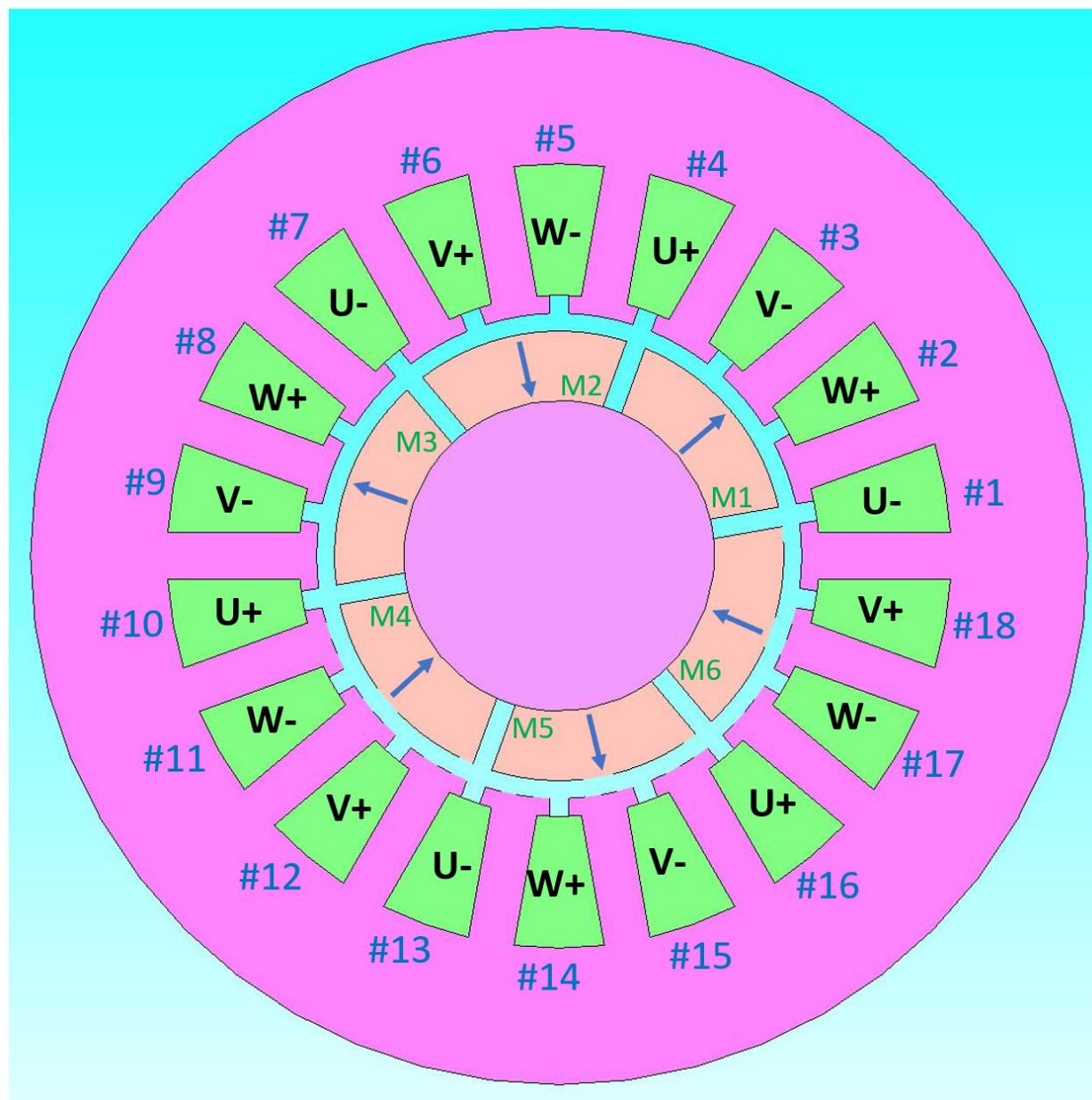
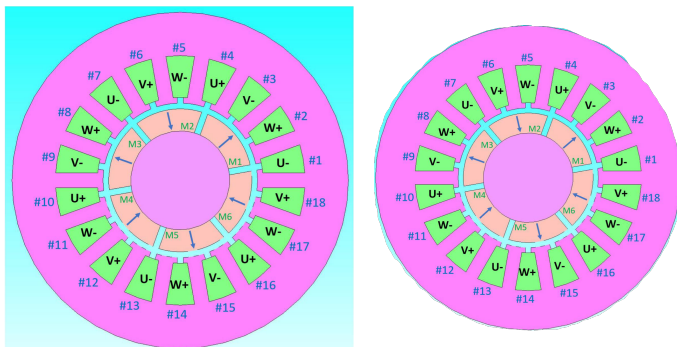
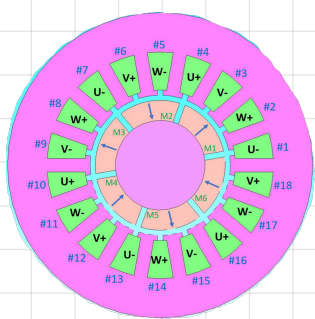




The figure above shows the cross section view of a surface permanent magnet synchronous motor at the initial rotor position. The motor has **18 slots** on the stator and **6 magnets** on the rotor. The coil directions for each phase at each slot are shown on the stator. The polarities of the magnets are also shown on the rotor. The rotor rotates in **counter clockwise direction**. Answer the following questions.

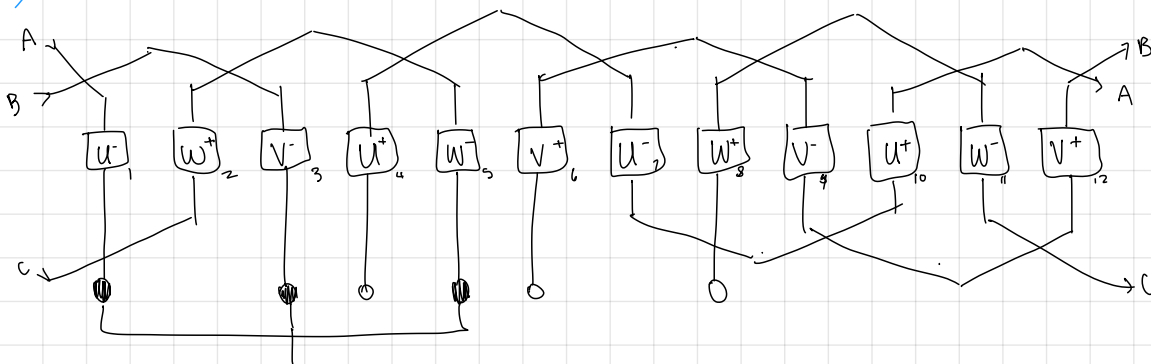
- Draw the **winding diagram** for this motor.
- Show the **magnetic poles** on the winding diagram when the phase currents with the following are applied: $i_u > 0$, $i_v < 0$, $i_w < 0$.
- Draw the no-load **phase flux linkage waveforms** for **one electrical cycle** as a function of **rotor position**. Clearly **show the rotor positions** when the phase flux linkage waveforms **cross zero**, and are at **positive and negative maximums**.
- Draw the no-load induced **phase voltage waveforms** for **one electrical cycle** as a function of **rotor position**. Assume sinusoidal waveforms. Clearly **show the rotor positions** when the induced phase voltage waveforms **cross zero**, and are at **positive and negative maximums**.
- For the **U-phase** only, draw the ideal sinusoidal **phase current waveform** as a function of **rotor position** that should be applied to generate **negative torque**.
- When the rotor rotates at **3000 rpm**, **calculate the mechanical frequency** of the rotor and the **electrical frequency** of the **current excitation**.





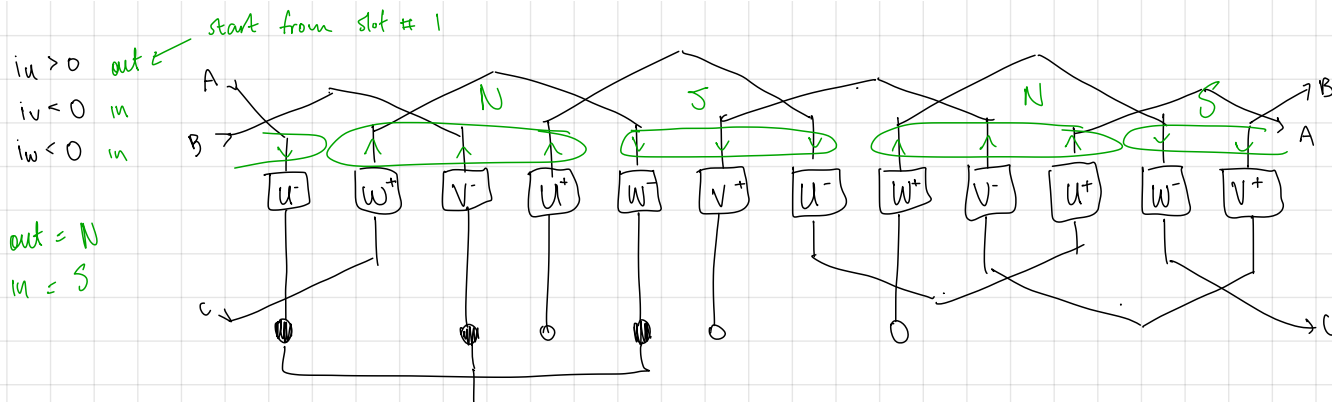
* forgot the last 6 slots
& the order is U, W, V not U, V, W...

a) (a) Draw the winding diagram for this motor.



I FORGOT
THE OTHER
6 Slots...

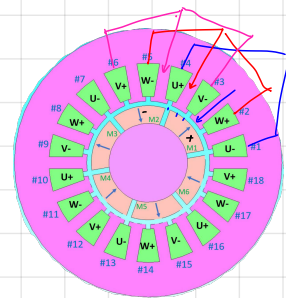
b) (b) Show the magnetic poles on the winding diagram when the phase currents with the following are applied: $i_u > 0$, $i_v < 0$, $i_w < 0$.



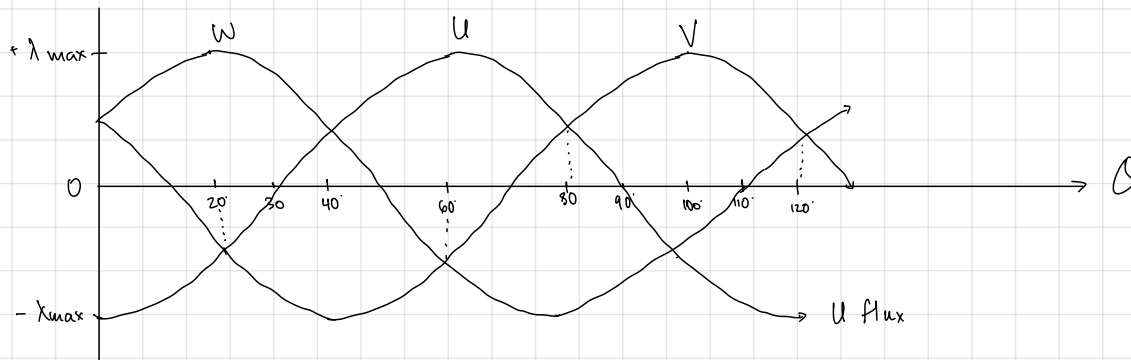
* there should be 6 magnets not 4... I forgot the other 6 slots..

* $U^+ \times +i = \text{out}$
 $U^- \times -i = \text{out}$
 $U^+ \times -i = \text{in}$
 $U^- \times +i = \text{in}$

c) Draw the no-load **phase flux linkage** waveforms for **one electrical cycle** as a function of **rotor position**. Clearly **show the rotor positions** when the phase flux linkage waveforms **cross zero**, and are at **positive and negative maximums**.



- elec cycle = $\frac{360}{PP} = \frac{360}{6/2} = \frac{360}{3} = 120^\circ$
- phase offset = $\frac{\text{elec cycle}}{PP} = \frac{120}{3} = 40^\circ$
- currently aligned: U slots w/ opposing directions $\rightarrow -\lambda_{\max} @ \theta = 0$
- important cross ($\pm \lambda_{\max}, 0$): $\frac{\text{phase offset}}{4} = 10^\circ \rightarrow 0^\circ, 10^\circ, 20^\circ, 30^\circ, 40^\circ$



d) Draw the no-load induced **phase voltage waveforms** for **one electrical cycle** as a function of **rotor position**. Assume sinusoidal waveforms. Clearly **show the rotor positions** when the induced phase voltage waveforms **cross zero**, and are at **positive and negative maximums**.

$$\mathcal{E} = \frac{d\lambda}{dt} \rightarrow \text{take derivative of each wave form above}$$



e) For the **U-phase** only, draw the ideal sinusoidal **phase current waveform** as a function of **rotor position** that should be applied to generate **negative torque**.

- $\hookrightarrow -T \rightarrow$ want I in opposite direction
- $\hookrightarrow \therefore$ just a -ve waveform of U's voltage waveform

f) When the rotor rotates at **3000 rpm**, calculate the **mechanical frequency** of the rotor and the **electrical frequency** of the **current excitation**.

$$W = 3000 \text{ rpm}$$

$$f_{\text{elec}} = ?$$

$$f_{\text{mech}} = ?$$

$$T_{\text{mech}} = 360^\circ = 2\pi$$

$$T_{\text{elec}} = 120^\circ = \frac{2}{3}\pi$$

$$\frac{3000 \text{ revs}}{1 \text{ min}} = \frac{3000 \text{ revs}}{1 \text{ min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot \frac{2\pi \text{ rad}}{\text{revs}} = \frac{3000 \cdot 2\pi}{60} = 100\pi \text{ rad/s}$$

$$f_{\text{mech}} = 100\pi \text{ rad/s}$$

$$f_{\text{elec}} = 100\pi \cdot \frac{360}{120} = 300\pi \text{ rad/s}$$

$$T_m = \frac{3}{2} p \lambda_{pm} i_q$$

$$= J \omega' + B \omega + T_L$$

$$\mathcal{E} = \lambda_{pm} \omega p$$

1. A 4-pole permanent magnet synchronous motor is rotated at 1500 rpm at no-load condition and the peak value of the phase voltages were measured as 30 V. The no-load phase voltage and no-load flux linkage waveforms have ideal sinusoidal shape. In order to generate 0.5 Nm of torque, calculate the peak value of the phase currents that should be applied to this motor. What should be the phase shift angle between the no-load phase voltage waveforms and the phase current waveforms?

2. A 6-pole permanent magnet synchronous motor is rotated at 3000 rpm at no-load condition and the peak value of the phase voltages were measured as 120 V. The no-load phase voltage and no-load flux linkage waveforms have ideal sinusoidal shape. In order to generate 2 Nm of torque, calculate the peak value of the phase currents that should be applied to this motor. What should be the phase shift angle between the no-load phase voltage waveforms and the phase current waveforms?

3. A 4-pole three-phase permanent magnet synchronous motor produces 1.5 Nm torque when phase currents with a magnitude of 10 A are applied to the motor in phase with the no-load phase voltage. Calculate the magnitude of the no-load phase voltage when the rotor is rotated at 3000 rpm. The no-load phase voltage and no-load flux linkage waveforms have ideal sinusoidal shapes.

- Q) 1. A 4-pole permanent magnet synchronous motor is rotated at 1500 rpm at no-load condition and the peak value of the phase voltages were measured as 30 V. The no-load phase voltage and no-load flux linkage waveforms have ideal sinusoidal shape. In order to generate 0.5 Nm of torque, calculate the peak value of the phase currents that should be applied to this motor. What should be the phase shift angle between the no-load phase voltage waveforms and the phase current waveforms?

$$T_m = \frac{3}{2} p p \lambda_{pm} i_q$$

$$= J \omega' + B \omega + T_L$$

$$\mathcal{E} = \lambda_{pm} \omega p p$$

$$i_q = ?$$

$$\theta_{shift} = ?$$

$$p p = \frac{4}{2} = 2$$

$$\omega = 1500 \text{ rpm} \cdot \frac{2\pi}{60} = 50\pi \text{ rad/s}$$

$$\mathcal{E} = 30 \text{ V}$$

$$T_m = 0.5$$

$$T_m = \frac{3}{2} p p \lambda_{pm} i_q$$

$$0.5 = \frac{3}{2} (2) \lambda_{pm} i_q$$

$$0.5 = 3 \lambda_{pm} i_q$$

$$0.5 = 3(0.095) i_q$$

$$i_q = 1.75 \text{ A}$$

$$\mathcal{E} = \lambda_{pm} \omega p p$$

$$30 = \lambda_{pm} (50\pi) (2)$$

$$30 = 100\pi \lambda_{pm}$$

$$\lambda_{pm} = 0.095$$

- 2) 2. A 6-pole permanent magnet synchronous motor is rotated at 3000 rpm at no-load condition and the peak value of the phase voltages were measured as 120 V. The no-load phase voltage and no-load flux linkage waveforms have ideal sinusoidal shape. In order to generate 2 Nm of torque, calculate the peak value of the phase currents that should be applied to this motor. What should be the phase shift angle between the no-load phase voltage waveforms and the phase current waveforms?

- 3) 3. A 4-pole three-phase permanent magnet synchronous motor produces 1.5 Nm torque when phase currents with a magnitude of 10 A are applied to the motor in phase with the no-load phase voltage. Calculate the magnitude of the no-load phase voltage when the rotor is rotated at 3000 rpm. The no-load phase voltage and no-load flux linkage waveforms have ideal sinusoidal shapes.

$$T_m = \frac{3}{2} p p \lambda_{pm} i_q$$

$$1.5 = \frac{3}{2} \left(\frac{4}{2}\right) \lambda_{pm} (10)$$

$$1.5 = 30 \lambda_{pm}$$

$$\lambda_{pm} = 0.05$$

$$\mathcal{E} = \lambda_{pm} p p \omega$$

$$= 0.05 \left(\frac{4}{2}\right) 100\pi$$

$$\mathcal{E} = 10 \text{ V}$$

$$\omega = 3000 \text{ rpm} \cdot \frac{2\pi}{60} = 100\pi \text{ rad/s}$$